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MCMC identification of a parametric Wiener-AR system

Arbitrary model orders, without sparsity-promoting priors

1 The model and the inference objective

The aim is to infer the original linear-block, polynomial and output-noise coefficients, together with the innovation variance. The orders n_a, n_b, n_f, n_d are arbitrary symbols throughout: no numerical values or equality between them are assumed. For each identification they are specified and held fixed; selecting their values is a separate model-selection problem. We use proper Gaussian coefficient priors. This is a mathematical and computational design for an estimator; no MCMC identification run is reported here.

With the delay operator q^{-1}, write

$$A(q^{-1};a) = 1 + \sum_{i=1}^{n_a} a_i q^{-i}, \qquad B(q^{-1};b) = \sum_{i=1}^{n_b} b_i q^{-i},$$
$$D(q^{-1};d) = 1 + \sum_{i=1}^{n_d} d_i q^{-i}. \tag{1}$$

The Wiener system and its autoregressive output noise are

$$\alpha_t = \frac{B(q^{-1};b)}{A(q^{-1};a)} u_t, \qquad \beta_t = \alpha_t + \sum_{j=2}^{n_f} f_j \alpha_t^j,$$
$$c_t = \beta_t + e_t, \qquad D(q^{-1};d)e_t = \nu_t, \quad \nu_t \overset{\text{iid}}{\sim} \mathcal{N}(0,s), \qquad s = \sigma_\nu^2 > 0. \tag{2}$$

Inputs and innovations are assumed independent. There is no process noise inside the linear signal block. Consequently, α and β are deterministic once the coefficients, input history and initial signal state have been specified. Only the additive output-noise path is stochastic.

Keep the normalization $f_1 = 1$ and no constant polynomial term, as in the manuscript. The physical coefficient vector is

$$\theta = (a_1, \ldots, a_{n_a}, b_1, \ldots, b_{n_b}, f_2, \ldots, f_{n_f}, d_1, \ldots, d_{n_d})^\top.$$

The normalization removes one scale ambiguity, assuming the true linear polynomial coefficient is nonzero. Suitable excitation and a nonredundant model are still needed; normalization alone does not prove identifiability.

1.1 Dimensions and recursions for arbitrary orders

There are $n_\theta = n_a + n_b + (n_f - 1) + n_d$ free coefficients and one variance. Thus the joint sampling dimension is $n_\theta + 1$, before adding unknown initial states. Here n_f is the polynomial degree, while $n_f - 1$ coefficients are estimated because $f_1 = 1$. The direct recursions are

$$\alpha_t = -\sum_{i=1}^{n_a} a_i \alpha_{t-i} + \sum_{i=1}^{n_b} b_i u_{t-i},$$
$$e_t = -\sum_{i=1}^{n_d} d_i e_{t-i} + \nu_t. \tag{3}$$

Empty sums are zero: $n_a = 0$ gives an FIR signal block and $n_d = 0$ gives white output noise. If $n_f = 1$, there are no free nonlinear coefficients.

For one identification, every chain and every posterior draw uses the **same observed record** $\mathcal{D} = (u, c)$. MCMC iterations generate parameter draws; they do not regenerate the input or the measured output. Physical time t and MCMC iteration k are different indices.

2 Specify initial conditions before writing the likelihood

Initial conditions are part of the statistical model. They determine the first samples of the predicted signal and the probability assigned to the beginning of the noise record. Signal and noise initial conditions require separate choices.

2.1 Initial conditions of the deterministic signal block

For each proposed (a, b), simulate the IIR block using one of the following explicit models.

1. **Known start and available input history.** Start from the known state before the experiment, propagate through all available input prehistory, and evaluate the likelihood only on the retained measured samples.
2. **Unknown state at the beginning of the retained record.** Use a state-space realization of B/A, introduce its initial state x_0, and give it a proper Gaussian prior, for example $x_0 \sim \mathcal{N}(0, V_x)$. Sample it together with (θ, s). The prior scale depends on the chosen state coordinates.

A state-space initial state can absorb the effect of unobserved past inputs. If a direct difference equation is used instead, the required past outputs and past inputs must be supplied consistently.

The existing study generator runs a burn-in period and then discards those samples. In general, the retained IIR trajectory begins with a nonzero state. Resetting that state to zero defines a different forward model. Its transient error may become small for long stable records, but that is an approximation, not an exact likelihood for the generated record.

2.2 Initial conditions of the AR output noise

Three common choices are legitimate, but they must not be mixed:

- **Stationary start:** the initial noise state follows the stationary distribution determined by (d, s). The main worked likelihood in this note uses this choice.
- **Known zero or known nonzero start:** condition on the specified noise state and propagate the innovations from that start.
- **Conditional likelihood:** condition on an initial block of observations and omit its density. This uses fewer likelihood contributions.

A finite burn-in from zero can approximate a stationary start. It is not identical to stationarity, particularly for poles close to the unit circle. For an exact simulation benchmark, match the likelihood to the actual generator.

2.3 General Gaussian form

Let $e = c - \beta(\theta)$, suppressing a known or sampled x_0. For a zero-mean noise initialization,

$$c \mid u, \theta, s \sim \mathcal{N}(\beta(\theta), sK(d)), \qquad Q = e^\top K(d)^{-1} e. \tag{4}$$

Hence

$$\log L = -\frac{N}{2}\log(2\pi s) - \frac{1}{2}\log|K(d)| - \frac{Q}{2s}. \tag{5}$$

For known zero noise prehistory, $K = HH^\top$ with a unit-diagonal lower triangular convolution matrix H, so $|K| = 1$; this finite-record covariance is generally not Toeplitz. The stationary covariance is Toeplitz and includes uncertainty in the initial noise state. Evaluate quadratic forms by solves, never by explicitly forming K^{-1}. For a known nonzero initial noise state, add its propagated mean to β before forming the residual vector.

3 How the function $Q(\theta)$ is formed

Q measures the discrepancy between the observed record and a proposed noise-free response, with the residual correlation accounted for. It is rebuilt at every proposed θ; it is not a fixed function of the observations alone. For clarity, the following formulas suppress a known or sampled signal initial state x_0. If it is unknown, write $Q(\theta, x_0)$ instead. Q is quadratic in the residual vector for fixed d, but is generally not quadratic in the full parameter vector θ.

3.1 Step 1: produce the proposed noise-free response

Use (a, b) and the fixed input to simulate the first recursion in (3), then apply the polynomial:

$$\beta_t(\theta) = \alpha_t(a, b) + \sum_{j=2}^{n_f} f_j \alpha_t(a, b)^j.$$

Changing a or b changes the full internal trajectory, not only one regression coefficient. Changing f changes the static transformation of it.

3.2 Step 2: subtract it from the observations

Form the output residual vector

$$e_t(\theta) = c_t - \beta_t(\theta), \qquad e(\theta) = (e_1(\theta), \ldots, e_N(\theta))^\top.$$

The unweighted sum $\sum_t e_t^2$ ignores serial correlation. For AR output noise, neighboring residuals must be interpreted jointly through d.

3.3 Step 3: whiten the residuals using the proposed noise model

At times with all required lags inside the record, apply $D(q^{-1}; d)$:

$$\begin{aligned} r_t(\theta) &= e_t(\theta) + \sum_{i=1}^{n_d} d_i e_{t-i}(\theta) \\ &= c_t - \beta_t(\theta) + \sum_{i=1}^{n_d} d_i [c_{t-i} - \beta_{t-i}(\theta)]. \end{aligned} \tag{6}$$

These are candidate innovation residuals. At the true coefficients and with the correct boundary treatment, they recover the driving innovations. Away from those coefficients they generally contain both model error and noise.

3.4 Step 4: sum squared whitened residuals, including the boundary

In the matrix formulation, factor $K(d) = L_d L_d^\top$ by Cholesky and solve $L_d w = e(\theta)$. Then

$$\boxed{Q(\theta) = e(\theta)^\top K(d)^{-1} e(\theta) = \|L_d^{-1} e(\theta)\|_2^2 = \sum_{t=1}^{N} w_t^2.} \tag{7}$$

The next section gives the same quantity using only a small initial block and the scalar residuals r_t, avoiding a dense $N \times N$ factorization.

$K(d)$ is the covariance for **unit innovation variance**. Consequently $Q(\theta)$ does not contain s: the full Gaussian exponent uses $Q(\theta)/(2s)$. Both the residuals (through a, b, f) and their whitening (through d) change with θ. For white noise, $K = I$ and $Q = \sum_t e_t^2$.

4 Compute Q and the likelihood for arbitrary AR order

For stationary output noise, define $\gamma_k(d) = \mathrm{Cov}(e_t, e_{t-k})/s$. These unit-variance autocovariances depend on d, not on s. For $n_d > 0$, the Yule-Walker equations are

$$\gamma_0 + \sum_{i=1}^{n_d} d_i \gamma_i = 1, \qquad \gamma_k + \sum_{i=1}^{n_d} d_i \gamma_{|k-i|} = 0, \quad k = 1, \dots, n_d. \tag{8}$$

Solve for $\gamma_0, \dots, \gamma_{n_d}$. Let $n_0 = \min(n_d, N)$ and form the initial-block covariance

$$P_d = [\gamma_{|i-j|}(d)]_{i,j=1}^{n_0}, \qquad e_{1:n_0} \sim \mathcal{N}(0, sP_d). \tag{9}$$

A small state-space Lyapunov equation is an equivalent way to compute these covariances. They must be recomputed at each proposed d.

4.1 Stationary initial block plus subsequent innovations

Factor the residual density into its initial block and any subsequent AR transition densities. With $P_d = L_P L_P^\top$, this yields

$$\boxed{\begin{aligned} Q(\theta) &= \underbrace{\|L_P^{-1} e_{1:n_0}(\theta)\|_2^2}_{\text{initial-block contribution}} + \underbrace{\sum_{t=n_0+1}^{N} r_t(\theta)^2}_{\text{innovation contribution}}, \\ \log L(\theta, s) &= -\frac{N}{2}\log(2\pi s) - \frac{1}{2}\log|P_d| - \frac{Q(\theta)}{2s}. \end{aligned}} \tag{10}$$

The first term assigns the correct probability to the beginning of the record, whose missing earlier noise samples are integrated out. Simply setting them to zero would give a different Q. The transformation from e to $(e_{1:n_0}, r_{n_0+1:N})$ is triangular with unit diagonal. Thus $|K(d)| = |P_d|$, and (10) equals the matrix Gaussian likelihood.

Compute the first contribution by solving $L_P w_0 = e_{1:n_0}$ and taking $w_0^\top w_0$. Then accumulate r_t^2 sequentially. Also compute $\log|P_d| = 2\sum_i \log(L_P)_{ii}$. A fixed small AR order gives a cost linear in N, plus the small covariance calculation.

4.2 Other boundary conventions

With **known zero noise prehistory**, set $e_t = 0$ for $t \le 0$, form $r_t = e_t + \sum_i d_i e_{t-i}$ for every $t = 1, \dots, N$, and use $Q = \sum_{t=1}^{N} r_t^2$. The log determinant is zero. With a **conditional** likelihood given the first n_0 outputs, omit the initial-block contribution, use $Q_{\mathrm{cond}} = \sum_{t=n_0+1}^{N} r_t^2$, and replace N by $N - n_0$ in the Gaussian normalization. Neither choice is the full stationary likelihood.

For $n_d = 0$, the initial block is empty, its contribution and log determinant are zero, and $r_t = e_t$. If $N \le n_d$, the initial block contains the whole record and the subsequent-innovation sum is empty.

> Q is the squared-error part of the likelihood, not the whole likelihood. When d and s are unknown, retain both the covariance determinant and the $-\frac{N}{2}\log s$ term. Differentiation must include the dependence of P_d on d and of e on the signal-path coefficients.

5 Reasonable priors without sparsity promotion

Choose proper, fixed covariance matrices, in the physical units of the coefficients. A starting family is

$$\begin{aligned} p(a) &\propto \mathcal{N}(a; 0, V_a)\, \mathbf{1}\{A \text{ is stable}\}, & b &\sim \mathcal{N}(0, V_b), \\ f &\sim \mathcal{N}(0, V_f), & p(d) &\propto \mathcal{N}(d; 0, V_d)\, \mathbf{1}\{D \text{ is stable}\}, \\ s &\sim \mathrm{IG}(\xi_0, \zeta_0), & \xi_0, \zeta_0 &> 0. \end{aligned} \tag{11}$$

Here $f = (f_2, \ldots, f_{n_f})^\top$. These parameter blocks are independent *a priori*, including independence from s. Within a stability-truncated block, the coefficients need not be independent even if the underlying Gaussian covariance is diagonal. Nonzero prior means can be used when justified. Empty parameter blocks are omitted, with zero contribution to the log prior.

The variance convention is

$$p(s) = \frac{\zeta_0^{\xi_0}}{\Gamma(\xi_0)} s^{-\xi_0 - 1} \exp(-\zeta_0/s), \quad s > 0. \tag{12}$$

Equivalently, the precision $\tau = 1/s$ has a Gamma prior with shape ξ_0 and **rate** ζ_0. A Gamma prior directly on s is also valid, but does not give the usual conjugate Gaussian variance update.

5.1 An illustrative calibration, not a universal default

For arbitrary block lengths, possible covariance families are

$$V_a = \tau_a^2 I_{n_a}, \quad V_b = \tau_b^2 I_{n_b}, \quad V_f = \mathrm{diag}(\tau_{f,2}^2, \ldots, \tau_{f,n_f}^2), \quad V_d = \tau_d^2 I_{n_d},$$

with fixed positive scales chosen in the coefficient units. The Gaussian densities for a and d remain restricted to stable filters; individual coefficient bounds alone do not enforce higher-order stability. Distinct polynomial scales can reflect different plausible contributions of different powers over the expected input range. For the variance, one possible choice is

$$s \sim \mathrm{IG}(3, 2\sigma_{\mathrm{ref}}^2), \qquad \mathbb{E}[s] = \sigma_{\mathrm{ref}}^2, \quad \mathrm{SD}(s) = \sigma_{\mathrm{ref}}^2.$$

Choose σ_{ref} from independent calibration, measurement specifications or a defensible noise range. Do not use the hidden true variance of a simulation as information available to the estimator. These scales should be revised if their implied outputs are implausible.

Draw coefficients and variances from the prior, simulate responses for the experimental input, and inspect amplitudes, transients and noise correlation. This prior predictive check is more informative than inspecting coefficient standard deviations in isolation. Stable poles can still imply very large gains.

5.2 Units and interpretation

Fixed Gaussian priors regularize the fit but do not select a sparse model: no coefficient is pruned and no coefficient-specific precision is learned. An excessively tight prior can materially affect finite-sample estimates.

If rescaling is useful, preserve the normalization. Under $u' = u/s_u$, $c' = c/s_c$ and $\alpha' = \alpha/s_c$,

$$a' = a, \quad d' = d, \quad b' = \frac{s_u}{s_c} b, \quad f_j' = f_j s_c^{j-1}, \quad s' = s/s_c^2.$$

Transform priors consistently. Arbitrarily subtracting the output mean is not innocuous when the static model has no constant term.

6 Enforce stability with smooth sampling coordinates

This section separates three questions: which physical parameters are allowed, how an unrestricted numerical vector represents them, and how their probability density must change when the coordinates change. The model orders remain arbitrary throughout.

6.1 Physical parameters and numerical coordinates are different objects

The physical unknowns are $x = (\theta, s)$: the coefficients in the Wiener-AR equations and the innovation variance. The prior already restricts a and d to stable denominator polynomials and requires $s > 0$. We now seek a smooth, one-to-one map $x = g(z)$ from unrestricted real coordinates z onto that domain. The transformation implements the chosen domain; it does not add parameters or change the forward equations.

For a generic denominator with m free coefficients, write

$$H(q^{-1}; h) = 1 + \sum_{j=1}^{m} h_j q^{-j}, \qquad C_m(\lambda) = \lambda^m + h_1 \lambda^{m-1} + \cdots + h_m.$$

Stability means that the roots of C_m lie strictly inside the unit circle. The coefficients h_j are coupled by this condition. Restricting each h_j separately to $(-1, 1)$ does not, in general, describe the stable region. We use $m = n_a$ and $h = a$ for the signal denominator, then $m = n_d$ and $h = d$ for the noise denominator. Neither order is assigned a numerical value.

6.2 First map: real coordinates to reflection coefficients

Introduce an auxiliary vector $\kappa = (\kappa_1, \ldots, \kappa_m)^\top$. These reflection coefficients are coordinates for a stable polynomial; they are generally different from its physical coefficients h. For every real $z_{c,i}$, set

$$\kappa_i = \tanh z_{c,i} = \frac{\exp(2z_{c,i}) - 1}{\exp(2z_{c,i}) + 1} \in (-1, 1), \qquad \frac{\partial \kappa_i}{\partial z_{c,i}} = 1 - \kappa_i^2 > 0. \tag{13}$$

The inverse is

$$z_{c,i} = \operatorname{atanh} \kappa_i = \frac{1}{2} \log \frac{1 + \kappa_i}{1 - \kappa_i}.$$

Thus every interior value $\kappa_i \in (-1, 1)$ corresponds to exactly one finite real coordinate. No clipping or boundary rejection is needed to impose these interval constraints mathematically. The next map turns κ into h.

The positive innovation variance uses a separate scalar transformation:

$$s = \exp v, \qquad v = \log s \in \mathbb{R}, \qquad \frac{ds}{dv} = \exp v = s. \tag{14}$$

The numerator coefficients b and polynomial coefficients f stay in their original coordinates; their transformation matrices are identities.

> The numerical sampler will move in z. At every proposed z, we recover the physical (a, b, f, d, s) and evaluate the same Wiener-AR model and physical priors. The maps tanh and exp are computational choices; the prior on the physical parameters remains the one specified in Section 5.

6.3 Second map: reflection coefficients to the physical polynomial

The Schur/Levinson step-up recursion builds the denominator one order at a time [7]. Here r is the intermediate polynomial order, not a data-time index or an MCMC iteration. The superscript in $h_j^{(r)}$ denotes that order, not a power.

Start with an empty coefficient vector $h^{(0)}$. For each $r = 1, \ldots, m$, take a copy of all coefficients at order $r-1$ and compute

$$\begin{aligned} h_r^{(r)} &= \kappa_r, \\ h_j^{(r)} &= h_j^{(r-1)} + \kappa_r h_{r-j}^{(r-1)}, \qquad j = 1, \ldots, r-1. \end{aligned} \tag{15}$$

The new coefficient is κ_r. Each earlier coefficient receives a correction equal to κ_r times the coefficient at its reversed position in the *previous* vector. All right-hand sides use that previous vector, so an implementation must not overwrite it partway through the step.

After the final step, $h = h^{(m)}$; denote this result by $\mathcal{T}_m(\kappa)$. The real Schur stability criterion states that the resulting polynomial is strictly stable exactly when all its reflection coefficients satisfy $|\kappa_i| < 1$ [7]. This is the reason for bounding κ, rather than independently bounding the original polynomial coefficients.

6.4 Why the map can be inverted

At a generic recursion step, a reversed pair obeys

$$h_j^{(r)} = u_j + \kappa_r u_{r-j}, \qquad h_{r-j}^{(r)} = u_{r-j} + \kappa_r u_j, \qquad u = h^{(r-1)}.$$

Subtract κ_r times the second equality from the first. The terms in u_{r-j} cancel, leaving $(1-\kappa_r^2)u_j$. Hence the step-down inverse is

$$\kappa_r = h_r^{(r)}, \qquad h_j^{(r-1)} = \frac{h_j^{(r)} - \kappa_r h_{r-j}^{(r)}}{1-\kappa_r^2}, \quad j = 1, \ldots, r-1. \tag{16}$$

Since $|\kappa_r| < 1$, the denominator is positive. Repeating backwards recovers all reflection coefficients, then atanh recovers z_c. This explains the one-to-one correspondence in the interior of the stable domain.

6.5 Assemble the complete coordinate map

Apply the same generic construction independently to a and d:

$$a = \mathcal{T}_{n_a}(\tanh z_a), \qquad d = \mathcal{T}_{n_d}(\tanh z_d), \qquad s = \exp v. \tag{17}$$

The sampling vector is

$$z = (z_a^\top, b^\top, f^\top, z_d^\top, v)^\top \in \mathbb{R}^{n_\theta + 1}.$$

Unknown signal initial states can be appended in their original real coordinates. If a denominator has order zero, its coordinate block and recursion are empty. The number of unknowns is unchanged by this transformation.

6.6 Why a change of coordinates requires a density correction

Let $\pi_x(x) = p(\theta, s \mid \mathcal{D})$ denote the intended posterior density with respect to physical-coordinate volume dx. Consider a small region around z. Its image under $x = g(z)$ has volume

$$dx \approx |\det Dg(z)|\, dz, \qquad Dg(z) = \frac{\partial x}{\partial z}.$$

The same event must have the same probability in either set of coordinates:

$$\begin{aligned} \pi_z(z)\, dz &\approx \pi_x(g(z))\, dx \\ &\approx \pi_x(g(z))\, |\det Dg(z)|\, dz. \end{aligned}$$

Dividing by dz and taking the limit gives the change-of-variables formula [2]:

$$\boxed{\pi_z(z) = \pi_x(g(z))\, J(z), \qquad J(z) = |\det Dg(z)|.} \tag{18}$$

The factor is multiplied because we used the derivative of the map *from sampling coordinates to physical coordinates*. Using the inverse derivative would give the reciprocal factor. In log form,

$$\log \pi_z(z) = \log \pi_x(g(z)) + \log J(z).$$

The probability distribution is preserved, although its density height changes with the local volume scale. If one samples $\pi_x(g(z))$ without the factor, the mapped physical draws instead have density proportional to $\pi_x(x)/J(g^{-1}(x))$. The missing factor therefore changes the answer.

6.7 A scalar illustration of the direction of the correction

Consider one auxiliary scalar κ with a uniform density $1/2$ on $(-1, 1)$. This is an illustration of coordinate volume, not a proposed prior for the Wiener coefficients. If $\kappa = \tanh z_c$, then

$$p_{z_c}(z_c) = \frac{1}{2}(1 - \tanh^2 z_c).$$

Indeed, its integral over $\mathbb{R}$ is $\frac{1}{2}[\tanh z_c]_{-\infty}^{\infty} = 1$. Equal-length intervals far from zero map to shorter intervals near $\kappa = \pm 1$, so they must receive less probability. Omitting the derivative would leave a constant over $\mathbb{R}$, which cannot even be normalized.

6.8 A worked correction for the innovation-variance prior

The prior is $p_s(s) \propto s^{-\xi_0 - 1} \exp(-\zeta_0/s)$. Substitute $s = e^v$ and multiply by $ds/dv = e^v$:

$$\begin{aligned} p_v(v) = p_s(e^v)e^v &\propto e^{-(\xi_0+1)v} \exp(-\zeta_0 e^{-v})e^v, \\ \log p_v(v) &= \text{const} - \xi_0 v - \zeta_0 e^{-v}. \end{aligned} \tag{19}$$

Thus the extra $+v$ removes one power of s from the density expression. The physical prior remains inverse-gamma; its density in the log-variance coordinate has a different functional form.

6.9 Derive the Jacobian of one denominator map, step by step

The denominator map is the composition $z_c \mapsto \kappa \mapsto h$. By the chain rule for matrices and multiplicativity of determinants,

$$\mathcal{J}_m(z_c) = \left|\det \frac{\partial h}{\partial \kappa}\right| \left|\det \frac{\partial \kappa}{\partial z_c}\right|. \tag{20}$$

The first coordinate map. Each κ_i depends only on $z_{c,i}$, so (13) gives a diagonal matrix and its determinant:

$$\frac{\partial \kappa}{\partial z_c} = \operatorname{diag}(1-\kappa_1^2, \ldots, 1-\kappa_m^2), \qquad \det \frac{\partial \kappa}{\partial z_c} = \prod_{i=1}^{m}(1-\kappa_i^2).$$

One recursion step. Put $u = h^{(r-1)}$ and let R_{r-1} reverse the order of its entries. Equation (15) can be written

$$h^{(r)} = \begin{pmatrix} (I_{r-1} + \kappa_r R_{r-1})u \\ \kappa_r \end{pmatrix}.$$

Differentiate with respect to the input vector $(u^\top, \kappa_r)^\top$:

$$\frac{\partial h^{(r)}}{\partial (u, \kappa_r)} = \begin{pmatrix} I_{r-1} + \kappa_r R_{r-1} & R_{r-1}u \\ 0^\top & 1 \end{pmatrix}. \tag{21}$$

It is block triangular. Its determinant is therefore $\det(I_{r-1} + \kappa_r R_{r-1})$; the column $R_{r-1}u$ does not affect it.

Evaluate that determinant. Reversing twice returns the original vector, so $R_{r-1}^2 = I$. Its eigenvectors can be symmetric or antisymmetric under reversal. Each mirrored pair of coordinates gives one of each kind; if there is an unpaired middle coordinate, it is symmetric. Consequently there are $\lceil (r-1)/2 \rceil$ eigenvalues $+1$ and $\lfloor (r-1)/2 \rfloor$ eigenvalues -1. The eigenvalues of $I + \kappa_r R$ are thus $1 + \kappa_r$ and $1 - \kappa_r$, giving

$$\det(I_{r-1} + \kappa_r R_{r-1}) = (1+\kappa_r)^{\lceil (r-1)/2 \rceil}(1-\kappa_r)^{\lfloor (r-1)/2 \rfloor}. \tag{22}$$

Both bases are positive because $|\kappa_r| < 1$.

Compose all recursion steps. The order-one step is $h^{(1)} = \kappa_1$ and has determinant one. Each later step transforms the earlier coefficients and appends the next reflection coefficient. Repeated application of the chain rule multiplies their determinants. Combining with (20) yields

$$\mathcal{J}_m(z_c) = \prod_{i=1}^{m}(1-\kappa_i^2) \prod_{r=2}^{m}(1+\kappa_r)^{\lceil (r-1)/2 \rceil}(1-\kappa_r)^{\lfloor (r-1)/2 \rfloor}. \tag{23}$$

6.10 Combine the blocks and form the actual sampling target

Taking logarithms of (23) turns every product into a sum:

$$\begin{aligned}\log \mathcal{J}_m(z_c) = &\sum_{i=1}^{m} \log(1-\kappa_i^2) \\ &+ \sum_{r=2}^{m} \left[\left\lceil \frac{r-1}{2} \right\rceil \log(1+\kappa_r) + \left\lfloor \frac{r-1}{2} \right\rfloor \log(1-\kappa_r) \right].\end{aligned} \tag{24}$$

For $x = (a, b, f, d, s)$ and $z = (z_a, b, f, z_d, v)$, the full derivative is block diagonal: the a block depends only on z_a, the d block only on z_d, and s only on v. The b and f blocks are identities and have determinant one. Equation (14) supplies the remaining factor $s = e^v$. Therefore

$$\boxed{\log J(z) = \log \mathcal{J}_{n_a}(z_a) + \log \mathcal{J}_{n_d}(z_d) + v.} \tag{25}$$

An empty denominator block contributes zero to $\log J$: set $\mathcal{J}_0 = 1$. Unchanged real initial-state coordinates also contribute determinant one.

Let $\ell(\theta, s)$ be the physical log posterior, including the likelihood and the priors. The numerical target is

$$\widetilde{\ell}(z) = \ell(\theta(z), e^v) + \log \mathcal{J}_{n_a}(z_a) + \log \mathcal{J}_{n_d}(z_d) + v.$$

For the Gaussian coefficient priors and stationary-noise likelihood used here, the formula in the next section expands to

$$\begin{aligned}\widetilde{\ell}(z) = \text{ const} &- \left(\frac{N}{2} + \xi_0 \right) v - \left(\frac{Q(\theta(z))}{2} + \zeta_0 \right) e^{-v} \\ &- \frac{1}{2} \log |P_{d(z)}| - \frac{1}{2} R_{\theta(z)} + \log \mathcal{J}_{n_a}(z_a) + \log \mathcal{J}_{n_d}(z_d).\end{aligned} \tag{26}$$

Here R_θ is the Gaussian-prior quadratic form defined in the next section. The physical coefficient of $\log s$ is $-(N/2 + \xi_0 + 1)$; the variance Jacobian $+v$ changes it to $-(N/2 + \xi_0)$ above. Add an initial-state log prior if those states are unknown. The determinant of P_d belongs to the likelihood; it does not replace any factor in J.

6.11 What to compute, and when software does the correction

At each proposed z: compute both reflection vectors, run both recursions, set $s = e^v$, evaluate the physical likelihood and priors, evaluate the two log Jacobians, and add their sum and v to the physical log posterior. Differentiate that complete scalar target for HMC.

Automatic differentiation computes derivatives of the expressions supplied; it does not by itself insert a missing change-of-variables correction. A package can insert corrections for constraints it transforms internally. For example, if it handles a declared positive s through a log transform, its $+v$ is already included; do not add it twice. A user-defined recursion from z_a, z_d to a, d still needs its own correction [2]. These formulas assume the explicit tanh and exp maps above; use the actual map if it differs.

Assigning Gaussian priors directly to z_a, z_d is a different prior model. To preserve the physical Gaussian priors, evaluate those priors at $a(z_a)$ and $d(z_d)$ and retain the derived Jacobians. Finally, use stable log expressions near the boundaries and avoid clipping, which destroys the stated bijection.

7 The posterior target and one complete evaluation

Bayes' rule gives

$$p(\theta, s \mid \mathcal{D}) \propto L(\theta, s)\, p(a)p(b)p(f)p(d)p(s).$$

No evaluation of the evidence $p(\mathcal{D})$ is needed. With zero-mean Gaussian priors, define

$$R_\theta = a^\top V_a^{-1} a + b^\top V_b^{-1} b + f^\top V_f^{-1} f + d^\top V_d^{-1} d.$$

For the stationary AR noise model of arbitrary order, the log density with respect to physical coordinates (θ, s), up to parameter-independent constants, is

$$\boxed{\begin{aligned}\ell(\theta, s) = &-\left(\frac{N}{2} + \xi_0 + 1\right)\log s - \frac{Q(\theta)/2 + \zeta_0}{s} \\ &- \frac{1}{2}\log|P_d| - \frac{1}{2}R_\theta.\end{aligned}} \tag{27}$$

Outside the stable region or for $s \le 0$, the density is zero. Fixed prior normalizing constants can be omitted. If x_0 is unknown, use $Q(\theta, x_0)$ and add $\log p(x_0)$; do not silently fix it when sampling. For $n_d = 0$, interpret $\log|P_d|$ as zero. For a conditional likelihood, use Q_{cond} and $N - n_0$, and omit the initial-block determinant.

The numerical sampler targets the density in z coordinates:

$$\widetilde{\ell}(z) = \ell\big(\theta(z), s(z)\big) + \log J(z). \tag{28}$$

The stationary-noise determinant term in (27) and the change-of-variables term in (28) have different origins. Both are required for this model and this choice of sampling coordinates.

7.1 What the computer does at each proposed parameter vector

1. Transform z to (a, b, f, d, s) using (17).
2. Simulate α_t over the fixed input record, using the specified initial state and any available prehistory.
3. Form $\beta_t = \alpha_t + \sum_{j=2}^{n_f} f_j \alpha_t^j$ and $e_t = c_t - \beta_t$.
4. Whiten the residuals, include the chosen initial-noise contribution, and compute Q and the log likelihood.
5. Add coefficient and variance log priors, any initial-state prior, and the transformation log Jacobian.
6. Return $\widetilde{\ell}(z)$ and its gradient $\nabla_z \widetilde{\ell}(z)$ to the sampler.

Automatic differentiation can propagate through the deterministic recursion. No separate sampling of the entire α, β or innovation sequence is needed for this model. HMC requires first derivatives, not an inverse Hessian.

For a physical-model baseline, the exact IIR recursion is the clearest forward model. If an FSM approximation is used, its frequency grid, truncation and boundary convention become part of the likelihood approximation and must be controlled. The Bayesian calculation should not silently target a different forward model from the comparison methods.

8 How one Hamiltonian Monte Carlo transition works

MCMC constructs a correlated sequence whose stationary distribution is the posterior. Under appropriate convergence conditions, averages along that sequence approximate posterior expectations. It does not seek a sequence of ever smaller losses. A draw with a lower posterior density can be an entirely appropriate part of the sample.

For a concrete explanation, consider fixed-length HMC [3]. Let z_k be the current unconstrained position and let M be a positive definite momentum covariance, fixed during retained sampling. Introduce an auxiliary momentum p and define

$$p \sim \mathcal{N}(0, \mathsf{M}), \qquad H(z, p) = -\widetilde{\ell}(z) + \frac{1}{2} p^\top \mathsf{M}^{-1} p. \tag{29}$$

The momentum is an algorithmic variable, not a state or noise source of the Wiener system. Its normalization is constant for a fixed M.

8.1 Step 1: refresh the momentum

Draw a new p_0 and record $H_0 = H(z_k, p_0)$. This random refresh allows the chain to explore different directions across successive transitions.

8.2 Step 2: integrate a reversible proposal trajectory

Starting at (z_k, p_0), take L leapfrog steps of length ϵ. One step is

$$\begin{aligned} p_{1/2} &= p + \frac{\epsilon}{2} \nabla_z \widetilde{\ell}(z), \\ z' &= z + \epsilon \mathsf{M}^{-1} p_{1/2}, \\ p' &= p_{1/2} + \frac{\epsilon}{2} \nabla_z \widetilde{\ell}(z'). \end{aligned} \tag{30}$$

Each new gradient evaluation repeats the forward simulation and log-target calculation from Section 7. The leapfrog construction approximately conserves H and preserves volume. Reverse the final momentum to obtain the standard reversible proposal; this reversal leaves its kinetic energy unchanged.

8.3 Step 3: correct numerical integration error

For the proposal $(z_\star, p_\star)$, accept with probability

$$a_{\text{acc}} = \min\{1, \exp[H_0 - H(z_\star, p_\star)]\}. \tag{31}$$

Generate $U \sim \text{Uniform}(0, 1)$. If $U < a_{\text{acc}}$, set $z_{k+1} = z_\star$; otherwise set $z_{k+1} = z_k$. Discard the auxiliary momentum and refresh it at the next transition. A rejected transition still contributes a repeated position to the chain; removing such repetitions changes its law.

In exact arithmetic with a correct implementation, the Metropolis correction accounts for finite leapfrog error. Finite run length and poor exploration still cause Monte Carlo error. MCMC does not replace the posterior with a Gaussian approximation, and it does not guarantee that a short run has explored every posterior mode.

9 Use NUTS in practice and run several chains

Fixed-length HMC illustrates the mechanics, but choosing L by hand is inconvenient. A mature implementation of the No-U-Turn Sampler (NUTS) is a reasonable first implementation choice [1]. It builds a trajectory of candidate points and stops its expansion when continued movement begins to turn back. The implementation uses a valid candidate-selection rule over the trajectory; it is not simply fixed-length HMC with its last endpoint accepted by (31).

During warm-up, the implementation adapts the integration step size and a metric that helps handle different scales and correlations. Retained sampling then uses the adapted settings. Keep the package's divergence, trajectory length and acceptance diagnostics. A longer trajectory is more expensive than a short one, so MCMC iteration counts alone do not measure computation.

9.1 A concrete initial run protocol

The following counts are a starting experiment, not a convergence guarantee.

1. **Fix the data and statistical model.** Save the record, input history, model orders, initial-state treatment, priors and physical units. All chains must target exactly this same posterior.
2. **Choose dispersed plausible starting positions.** Use four different stable coefficient vectors and positive variances, converted to z coordinates. Check that every starting log target and gradient is finite. A fitted or MAP estimate can initialize one chain, but all chains should not be copies of the same point.
3. **Warm up each chain.** For example, use 1,000 warm-up transitions per chain. Let the sampler adapt and exclude these transitions from posterior summaries.
4. **Retain a first batch.** For example, retain 2,000 transitions per chain. This yields 8,000 raw draws across four chains, not 8,000 independent draws. Preserve chain identities for diagnostics.
5. **Assess exploration and precision.** Use the checks in Section 10. If exploration is sound but Monte Carlo errors remain too large, run longer. If there are unresolved divergences or modes, investigate the geometry, initialization or model before interpreting a longer run as a solution.
6. **Save enough to reproduce the result.** Store physical draws, sampler diagnostics, seeds, warm-up settings, software versions, elapsed time and the exact model and configuration used.

9.2 Two different meanings of burn-in

Data burn-in propagates the simulated physical filters before retaining the observation record. It affects the signal and noise initial distributions. **MCMC warm-up** adapts the sampler and moves chains away from their starting points. It does not modify the data or remove an incorrectly modeled physical transient. These two operations solve different problems.

No new innovation realizations are generated inside a likelihood evaluation. Randomness during sampling comes from the MCMC algorithm. New physical innovations are generated later for posterior predictive simulation.

10 Decide whether the posterior sample is trustworthy

Inspect the physical coefficients, s, any initial states, and important derived quantities. Trace and rank plots should show chains exploring the same distribution rather than remaining separated. Pair plots can reveal strong dependence, curved ridges or separate solutions.

Diagnostic	Interpretation and first action
Rank-normalized $\widehat{R}$	Compare within-chain and between-chain behavior. Aim for $\widehat{R} < 1.01$ for the reported quantities. Larger values call for investigation of poor mixing or different explored regions.
Bulk and tail ESS	Account for dependence between draws. With four chains, a bulk ESS above 400 is a baseline check, not a universal accuracy target. Tail ESS matters for interval endpoints and tail probabilities.
Divergences	Investigate every post-warm-up divergence before final inference. They can indicate that numerical trajectories fail in an important part of the posterior. Longer runs alone need not solve this.
Trajectory and energy checks	Repeated maximum-tree-depth events indicate costly exploration. Low energy-transition efficiency can reveal poor geometry. Check sampler warnings and parameter-pair plots.

These are practical screening rules, consistent with Stan's diagnostic guidance [4], not a proof that every possible mode has been discovered.

10.1 Quantify Monte Carlo error separately from posterior uncertainty

For a scalar quantity $g(\theta, s)$, the posterior standard deviation describes uncertainty about that quantity given the record. The Monte Carlo standard error (MCSE) describes numerical uncertainty in its estimated posterior mean. For a well-mixed stationary sample, a useful interpretation is

$$\mathrm{MCSE}(\overline{g}) \approx \frac{\mathrm{SD}_{\mathrm{post}}(g)}{\sqrt{\mathrm{ESS}_{\mathrm{mean}}(g)}}. \tag{32}$$

Use the software's appropriate mean-specific ESS or direct MCSE estimate; bulk ESS is a general diagnostic rather than exactly the same statistic [5].

For example, a desired MCSE below 2% of the posterior standard deviation suggests a mean-specific ESS of roughly 2,500. This illustrates why the baseline ESS of 400 need not suffice for precise numerical reporting. Quantile MCSE requires its own calculation.

10.2 If diagnostics are unsatisfactory

Check likelihood signs and initial conditions first. Then inspect parameter scales and posterior dependence, use realistic priors and improve smooth parameterizations. A smaller integration step may help some divergences, but does not cure nonidentifiability or guarantee movement between distant modes. If multiple modes persist, report the issue and consider methods designed for multimodal exploration. Do not pool chains stuck in different regions as though their relative frequencies were correct posterior weights.

11 Turn draws into parameter and prediction results

After satisfactory diagnostics, let $(\theta^{(m)}, s^{(m)})$, $m = 1, \dots, M$, be the retained physical draws, combining chains only for summaries. For every coefficient and the variance, report a posterior mean or median, posterior standard deviation and a 95% equal-tailed credible interval. Include a correlation plot when blocks compensate for each other.

$$\widehat{\theta}_{\mathrm{mean}} = \frac{1}{M} \sum_{m=1}^{M} \theta^{(m)}, \qquad I_j = [q_{0.025}(\theta_j), q_{0.975}(\theta_j)]. \tag{33}$$

Transform each draw before summarizing: in general $\mathbb{E}[\theta(z)] \neq \theta(\mathbb{E}[z])$, and $\mathbb{E}[\sqrt{s}] \neq \sqrt{\mathbb{E}[s]}$. If a physical-coordinate MAP is needed, maximize the physical density (27); maximizing the transformed density (28) defines a different MAP.

11.1 Noise-free output uncertainty

For each posterior draw, propagate the linear block and static nonlinearity to obtain $\beta_t^{(m)}$. Pointwise quantiles describe uncertainty about the noise-free response due to uncertain parameters and, where sampled, initial states. The posterior mean response is the average of these trajectories:

$$\widehat{\beta}_t = \frac{1}{M} \sum_{m} \beta_t^{(m)}.$$

It generally differs from evaluating the nonlinear forward model at $\widehat{\theta}_{\mathrm{mean}}$.

11.2 Predict a new noisy observation

For arbitrary AR order and a known next input, a one-step prediction conditional on an observed history containing the required lags is

$$\begin{aligned} e_{t+1-i}^{(m)} &= c_{t+1-i} - \beta_{t+1-i}^{(m)}, \quad i = 1, \dots, n_d, & \nu_{t+1}^{(m)} &\sim \mathcal{N}(0, s^{(m)}), \\ c_{t+1}^{(m)} &= \beta_{t+1}^{(m)} - \sum_{i=1}^{n_d} d_i^{(m)} e_{t+1-i}^{(m)} + \nu_{t+1}^{(m)}. \end{aligned} \tag{34}$$

These predictive intervals include both parameter uncertainty and new innovation noise, so they have a different interpretation from intervals for β. For several future steps, recursively propagate the AR noise and generate fresh innovations at each step. For an independent repeated record, also draw its initial noise state under the specified initialization model.

11.3 Interpretation for the manuscript's numerical studies

A credible interval is a posterior probability statement under the chosen model and prior; it is not automatically a frequentist coverage guarantee. Assess coverage, bias and RMSE by repeating identification on independently generated datasets. Within one dataset, MCMC draws measure posterior uncertainty and numerical integration error, not the sampling distribution across experiments.

Numerical inference does not intrinsically reduce statistical precision when it accurately explores the intended posterior. Precision still depends on record length, excitation, model correctness and prior information. Posterior means can have finite-sample bias, and a poorly mixed numerical approximation can introduce additional error.

12 Validate the implementation before a large study

Start with a short synthetic record and validate identities that have an independent answer. Knowing simulation truth is useful for debugging and later performance assessment; it is not information supplied to the estimator.

12.1 Likelihood and gradient checks

For stable coefficient vectors of several AR orders, compare the log density calculated from the dense covariance $sK(d)$ with (10). They must agree to numerical precision for the same residual vector and initial-state model. Separately check the physical-to-unconstrained Jacobian and compare automatic gradients with central finite differences at interior parameter points.

Check special cases: $d = 0$ reduces to independent output noise, $f_j = 0$ for $j \geq 2$ gives a linear signal path, and known zero noise prehistory gives the corresponding conditional innovation recursion. Include tests for the actual beginning-of-record convention, where indexing mistakes are common.

12.2 Two analytic conditional distributions for additional checks

Although the joint posterior for (a, b, f, d, s) is generally nonconjugate, some full conditional distributions are simple [6]. These identities also offer optional alternatives to sampling every parameter by HMC.

With θ and any signal initial state fixed, the inverse-gamma variance prior gives

$$s \mid \theta, \mathcal{D} \sim \mathrm{IG}\left(\xi_0 + \frac{N}{2},\ \zeta_0 + \frac{Q(\theta)}{2}\right). \tag{35}$$

Here N corresponds to the full stationary likelihood. Replace it by the number of conditional residual contributions if using a conditional likelihood. The independence of the coefficient priors from s matters to this formula.

Next fix (a, b, d, s) and the signal initial state. Define $Z_{t,j} = \alpha_t^{j+1}$, $j = 1, \ldots, n_f - 1$, and $w = c - \alpha$. With $f \sim \mathcal{N}(0, V_f)$,

$$\begin{aligned} S_f &= \left(V_f^{-1} + \frac{1}{s} Z^\top K(d)^{-1} Z\right)^{-1}, \\ m_f &= S_f \frac{1}{s} Z^\top K(d)^{-1} w, \qquad\qquad f \mid a, b, d, s, \mathcal{D} \sim \mathcal{N}(m_f, S_f). \end{aligned} \tag{36}$$

Use whitening or solves to implement these expressions. This is a Gaussian conditional for the polynomial coefficients, not a closed-form joint posterior for the rational and AR coefficients.

12.3 Optional Gibbs updates

One can alternate an exact draw from (35) with a valid MCMC update of the remaining parameters, or use (36) as another Gibbs block. The resulting hybrid sampler still needs convergence diagnostics and can mix poorly when blocks are strongly dependent. Joint NUTS for the complete (θ, s) vector remains a straightforward first baseline; analytic conditional updates are an optional refinement, not a prerequisite.

Finally, inspect posterior predictive behavior, whitened residual autocorrelation and sensitivity to reasonable prior scales. A single successful fit is not a finite-sample bias or coverage study.

13 Implementation order, computational cost and references

13.1 A practical order of work

1. Fix the forward model and initial conditions; implement and validate the scalar innovation likelihood and physical priors.
2. Add smooth stability transformations, their Jacobian and differentiable recursions; validate log targets and gradients.
3. Run a small four-chain NUTS pilot. Resolve diagnostic problems and quantify Monte Carlo errors before increasing record lengths or repetitions.
4. Save parameter summaries, credible intervals, predictions, diagnostics and computational cost for every dataset. Then add this estimator to the initialization and record-length experiments using the same observed records and explicitly chosen posterior point estimate.

For fixed orders, a forward likelihood evaluation is linear in record length, with approximate cost

$$O(N(n_a + n_b + n_f + n_d))$$

when evaluating the polynomial directly. Differentiation and repeated trajectory steps increase the constant and total work. A NUTS transition can require many full-record gradient evaluations. Record wall time, target/gradient evaluations and effective samples per second. Compare posterior-mean accuracy against GHAM and other estimators on the same datasets, while reporting the extra uncertainty information MCMC provides. Optimizer iterations and NUTS transitions are not comparable units of cost.

13.2 References and further reading

The likelihoods, stability map and posterior formulas above specialize standard Gaussian and MCMC calculations to the manuscript's Wiener-AR model. The following primary sources document the sampling machinery and diagnostics.

1. M. D. Hoffman and A. Gelman (2014). *The No-U-Turn Sampler: Adaptively Setting Path Lengths in Hamiltonian Monte Carlo.* Journal of Machine Learning Research 15(47), 1593–1623. Paper and full text.
2. Stan User's Guide. *Reparameterization and change of variables.* Official guide. See also the Reference Manual's constraint transforms. Transformation conventions must match the chosen implementation.
3. R. M. Neal (2011). *MCMC Using Hamiltonian Dynamics.* In Handbook of Markov Chain Monte Carlo, Chapter 5. Author manuscript.
4. Stan. *How to Diagnose and Resolve Convergence Problems.* Official diagnostic guidance. Practical thresholds are checks, not convergence proofs.
5. Stan Reference Manual. *Posterior Analysis.* ESS and Monte Carlo error.
6. K. P. Murphy (2007). *Conjugate Bayesian Analysis of the Gaussian Distribution.* Technical note. Gaussian variance and precision conventions; conditional conjugacy should not be confused with joint conjugacy for this nonlinear model.
7. MathWorks. *Reflection coefficients and prediction polynomials.* Step-up recursion and stability criterion. The Jacobian in this note follows by differentiating the stated recursion.

Sampling references checked on 29 October 2026; coordinate-transformation references rechecked on 30 October 2026. The illustrative run lengths and prior scales in this note are proposed starting choices, not empirical performance results.